Improving Text Readability to Support Student Comprehension and Learning: An LLM-Powered Approach

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Abstract. Reading proficiency is predictive of academic success, yet many students, especially those with diverse learning needs, struggle with complex academic texts. Existing support tools often fail to adequately address challenges related to the complexity of text vocabulary and grammar. However, large language models (LLMs) might be able to meet this need. We compared the effectiveness of two prompting strategies for simplifying academic texts (N = 2,000): one that used plain-text instructions and another that incorporated a readability metric. The Metric-Guided Prompt demonstrated a significant reduction in text complexity as measured by the Flesch-Kincaid Grade Level. Following this intrinsic evaluation, we conducted a between-subjects study with 37 students to determine whether there were differences in learner perceptions of the texts and their learning gains, based on the source of the information provided (i.e., the original and simplified texts). The results of both the intrinsic evaluation and the user study indicate that the Metric-Guided Prompt improves text readability without hindering learning. These findings underscore the potential for appropriately prompted LLMs to foster academic success for diverse learners by improving information access and supporting comprehension.

Keywords: Readability · Prompt Engineering · Learning · GenAI

1 Introduction

Reading is a fundamental skill that predicts student success [24]. Unfortunately, due to the complexity of academic texts, many students struggle to read and understand their course materials [26,32,37]. These difficulties are particularly common among students with diverse learning needs, including those with

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K. Tammets et al. (Eds.): EC-TEL 2025, LNCS 16063, pp. 291–305, 2026.
https://doi.org/10.1007/978-3-032-03870-8_20

dyslexia or attention deficit hyperactivity disorder (ADHD), or those who participate in educational programs offered in a non-native language [12]. Difficulties in reading comprehension, word reading, and spelling can hinder learning progress and academic achievement [20], limiting students' access to higher levels of education [24] and their future job opportunities [14]. The global decline in reading proficiency among students in the last decade, especially in recent years, further emphasizes the critical need to develop support in reading for all learners [19].

Existing mechanisms to support learners with weaker reading skills, such as text-to-speech tools, provide useful assistance for specific challenges that include difficulties in decoding written text and limited reading fluency [21]. They can also support general comprehension [5]. However, these tools often fail to help learners process the complex vocabulary and grammatical structures found in academic texts [32]. An obvious solution to this would be to simplify the grammar and vocabulary used in these texts. However, traditional methods for text simplification [6] can be labor intensive and may not accurately capture the essence of the original content [22], indicating a need for effective automated text simplification approaches.

Recent advances in natural language processing (NLP) and generative artificial intelligence (genAI) present promising opportunities to improve text readability. In particular, research indicates that large language models (LLMs), such as the one used in ChatGPT, can help reduce the complexity of academic texts through text simplification, improving readability based on established metrics like LIX, ('Läsbarhetsindex' in Swedish) [11]. However, despite the potential of these technological advances, their use in educational settings remains limited, and many students and educators are unaware of their potential benefits.

The current study builds on recent advances in NLP and preliminary findings suggesting the utility of LLMs in simplifying complex texts when appropriate prompting strategies are used [11,18]. Aiming to take advantage of the improved comprehension that simplified texts provide [4], we investigated the utility of including a *measure of text readability within the prompt*. We then evaluated how much students learned when reading the original text and when reading the simplified text generated by an LLM using the best-performing prompting strategy. This two-study approach provides insight into the ability of LLMs to make text accessible to students with varied learning needs by answering the below research questions:

RQ1: Does including a readability metric in the prompt support better text simplification?
RQ2: How readable do students feel the original and simplified text are?
RQ3: How much do students learn when reading the simplified text in contrast to the original text?

To answer these questions, we examine how generative AI can be used to automate the text simplification process and identify potential implications for educational practice and student learning.

2 Background

The challenges of reading academic texts often arise from complex sentences and advanced vocabulary [20,24,30]. Traditional readability metrics, such as the Flesch-Kincaid Grade Level (FKGL) [15] and the Gunning Fog Index [10], allow evaluation of key components of text readability using proxies for syntactic complexity and vocabulary difficulty [11]. Although these metrics can provide an estimate of how readable a text is (e.g., [9,13,28,31]), they fail to capture many of the aspects of a text that affect students' comprehension and retention of information [20]. This limited representation of the factors that contribute to reading comprehension implies that readability measures cannot be relied on as the only indicator of a text's readability. They can, however, be used as an initial proxy indicator when trying to increase the readability of texts through manual or automated means.

2.1 Improving Text Readability

Text readability can be increased through modifications that improve clarity and information access [7]. One such modification is text simplification, which typically involves simplifying the vocabulary and syntax within a text to reduce cognitive load and support comprehension [8]. Although text simplification can be beneficial for reading practices, there is a risk that essential information will be lost through the simplification process. When simplified texts are used for instructional purposes and learning, preserving key information and conceptual nuance is essential to avoid disadvantaging students who rely on this form of support. In short, the nuance contained in academic texts must be preserved as much as possible. Concerns about the potential loss of important information as part of the simplification process are probably the reason why early text simplification used manual methods to shorten the length of sentences, replace complex words with simpler synonyms, and restructure paragraphs [6,16,25]. Although effective, these manual methods are resource intensive and lack scalability.

While not targeted towards supporting student learning, NLP researchers have invested considerable effort in techniques for summarizing and simplifying text. The resulting approaches include rule-based transformations [27], machine-learning models that identify and rephrase difficult sentences [1], and the use of genAI [11]. These recent genAI approaches, especially LLMs, have surpassed previous techniques in their ability to transform complex text. Unlike earlier rule-based (e.g., [27]) or statistical methods (e.g., [33]), LLMs can adapt to diverse contexts to offer contextually appropriate transformations [3]. However, given the importance of maintaining key information and nuance in simplified texts, there remains a need to explore which prompting strategies most effectively support modifications that enhance the readability of academic texts.

2.2 Strategic Prompting for Improved Readability

Previous studies have begun to explore the potential of LLMs to enhance the readability of academic texts. In their recent study, Estling and colleagues [11]

examined this potential using 136 Swedish academic texts, while evaluating four prompting strategies: Standard, Meta, Roleplay, and Chain-of-Thought. The LIX readability metric [2] was included in all prompting strategies. It was also used as the target measure since the texts were in Swedish. The results showed Meta prompting best supported the task, but all prompting approaches produced more readable texts. This preliminary study only conducted an intrinsic evaluation of the simplification methods, as such, it is unclear how learners would respond to or learn from the simplified texts. This raises two questions, *What happens when a readability metric is not included in the prompt?* and *How do learners respond to LLM-simplified texts?*

Both of these questions need to be answered to determine the utility of LLMs to support student access to the information found in academic texts. Above and beyond examining the use of LLMs to improve readability from a metric perspective, the simplified texts also need to facilitate student learning. Therefore, it is necessary to evaluate this support with learners. In the present paper, we report the results of two studies. The first is the intrinsic evaluation that answers RQ1 by investigating how much the readability of the text changes using two prompting strategies: one that used plain-text instructions (Plain-Language Prompt) and another that incorporated a readability metric (Metric-Guided Prompt). Study 1's results aim to help identify which prompting strategy is more appropriate for improving text readability from a metric perspective. The second study then evaluates student learning and perceptions when they have access to either the simplified text or the original text (RQ2 & RQ3). Study 2's results aim to provide insights into how simplified text may support learners.

3 Study 1 - Intrinsic Evaluation

We conducted an intrinsic evaluation of text readability after they had been simplified using an LLM. We specifically compared the results of two prompting strategies to identify which prompting strategy should be used in the student reading study (see Sect. 4).

3.1 Methods

To enhance text readability, two prompting strategies, namely Metric-Guided Prompt and Plain-Language Prompt, were developed and tested. After simplifying the texts using these prompts, we obtained a readability measure for each of the original texts and the simplified versions that the LLM output in response to each of our two prompting strategies.

Prompting Procedures and Strategies. The GPT-3.5 Turbo LLM was used due to its affordability. This LLM employs two types of prompts. The first, a system prompt, determines what the LLM does in response to user prompts. This is the prompt we manipulated. The second type, the user prompt, provided the text that needed to be simplified.

The two system prompting strategies that we used were a *plain-language* prompt and a *metric-guided prompt* as this combination provides insight into the value of including the readability metric as part of the prompt.

Plain-Language Prompt: This approach broadly focused on text simplification by having the LLM rephrase sentences for clarity and replace complex vocabulary with simpler synonyms. It instructs the LLM to focus on improving readability while preserving the information contained in the original content. It is a flexible approach, which enables the model to simplify text without imposing rigid constraints.

The prompt used was the following: "You are a text enhancement AI. Your main goal is to make the text simpler while keeping the main idea. You will also correct any grammatical errors and improve the text. You will not change the main idea of the text. You will not add any new information to the text." This prompt is equivalent to the Standard prompt used in prior work [11] with one modification: it does not include the readability formula.

Metric-Guided Prompt: This strategy employed the FKGL measure of readability [15]. The use of this formula constrains the text simplification task so that it consistently targets the same aspects of the text. It emphasized measurable improvements in sentence structure and vocabulary complexity, with prompts explicitly instructing the model to optimize text based on predefined criteria. This prompt is equivalent to the previously-examined standard prompt strategy (see [11]). Additional information about how we refined the prompt can be seen in our repository.

The prompt used was the following: "Your task is to reduce the Flesch-Kincaid Grade Level of texts. The Flesch-Kincaid Grade Level is calculated using the formula: Grade Level = 0.39 * (Total Words/Total Sentences) + 11.8 * (Total Syllables/Total Words) - 15.59. Given a text, rewrite it to significantly reduce its complexity by focusing on: Simplifying sentence structures; Using fewer syllables per word; Reducing the average sentence length. Ensure the rewritten text maintains its original meaning while achieving a much lower grade level."

Both strategies were implemented through Python scripts[1], enabling automated processing of the texts and ensuring consistency in application.

Data. The effectiveness of the two prompts was evaluated using 2,000 academic papers from a dataset that contained 37,464 academic texts on artificial intelligence [34]. All texts were open-source academic preprints.

This dataset consists of four lists of files. We randomly sampled the employed texts from the first of the four files. We then cleaned the texts' introduction using a script that removed LaTeX artifacts. The text was retained if at least 50% of the original text remained following cleaning.

Each of the original texts that was sampled contained at least 46 words ($M = 793.2, SD = 619.36$) and four sentences ($M = 40.4, SD = 31.58$). The texts contained at most 5,813 words and 336 sentences.

[1] Repository: https://osf.io/ukcxf/?view_only=118f57f76bcd45b586ad049aed9c3393.

Readability Measure. We used the FKGL as a readability metric [15]. FKGL is commonly used in education, publishing, and software to evaluate the complexity of written material [29,31]. This measure uses information about sentence length and vocabulary complexity (number of syllables) to infer the approximate grade level associated with a text. Lower grade levels are considered to have higher readability [35].

Analysis. We used a repeated measures ANOVA to determine whether there were differences in FKGL across the three sets of texts (i.e., the original text and the two text versions that were generated using each prompting approach). Greenhouse-Geisser correction was used because the data did not meet the sphericity assumption as determined through Mauchly's tests. We applied Bonferroni correction to the post-hoc paired t-tests. The paired tests of raw scores were one-tailed because we expected the simplified text to have lower scores.

As others have done, we characterized the change in readability through an adaptation of the normalized gain score [11]. We used 22 (the ceiling of the maximum observed FKGL from our data) instead of the test score ceiling (i.e., 100) employed in the original formula. Because the gain data were not normally distributed, we used a Wilcoxon-signed rank test to determine whether there was a difference in the amount of improvement seen across prompting strategies. These pairwise tests were two-tailed because there was limited theoretical or empirical support for hypothesizing the direction of the difference.

3.2 Results

A large difference was found across versions of the texts: $F(1.76) = 241.54, p < .001, \eta^2 = .108$. The original texts had the highest grade level ($M = 13.1, SD = 2.14$), and only the *metric-guided prompting* strategy supported measurable changes in text-readability (Table 1). This prompting strategy also resulted in scores ($M = 12.1, SD = 2.38$) that were around one grade lower than those seen for the plain-language prompt ($M = 13.0, SD = 1.92$), and improvements over the original text could be as large as 13 grades for the metric-guided prompt ($M = 0.98, SD = 2.64$).

Table 1. Differences in FKGL across the original and simplified texts.

Text 1	Text 2	Statistics
Original	Metric-Guided	$t(1999) = 16.65, p < .001, d = 0.37$
Original	Plain-Language	$t(1999) = 1.04, p = .450, d = 0.23$
Plain-Language	Metric-Guided	$t(1999) = 20.12, p < .001, d = 0.45$

These differences are consistent with the changes seen in the number of sentences and words used to communicate information in the simplified texts

(Table 2). For both strategies, the median number of words is approximately one-third the median length of the original texts ($Mdn = 633$), and the number of sentences was half that of the original texts ($Mdn = 32$).

Table 2. Descriptive statistics of the simplified texts' characteristics (Mdn - median; IQR - inter-quartile range)

Strategy	Min.	Mdn	Max	IQR
Plain-Text Prompt				
Words	44	248	1440	103
Sentences	3	15	79	5
Metric-Guided Prompt				
Words	35	234	1135	117
Sentences	4	16	59	7

In addition to a difference in the FKGL scores, we found a medium difference in the amount of improvement seen when applying each prompting strategy ($Z = 17.99, p < .001, r = .40$). The apparent reductions in grade level map to negligible normalized gains ($M = 0.07, SD = 0.770$) for the plain text prompting strategy. The normalized gains for the metric-based prompting approach were three times higher ($M = 0.20, SD = 0.92$).

These results clearly indicate that the metric-based prompting strategy simplified the texts the most. We, therefore, used this strategy to simplify the texts for our follow-up study to determine whether this strategy could support student learning.

4 Study 2 – Learner Evaluation

This between-subjects study aimed to determine whether there were differences in student perceptions and learning based on the source of the information they were given. Two sources were considered: the original text and a simplified text. The simplified text was created using the best-performing approach (i.e., Metric-Guided Prompt) from Study 1.

4.1 Methods

We recruited participants through social-media groups for science, technology, engineering, and mathematics (STEM) students in November-December 2024. Participants were divided into two groups: one group read the original text ($n = 22$), while the other read the simplified text ($n = 15$). Participants were assigned to a group using a random number.

Participants. Participants ($n = 37$) were recruited based on their minimal prior knowledge of the academic topics used in the text. The STEM groups we advertised the study to were for students who did not focus on medicine or pharmaceuticals.

Of those who consented to participate, two were native-English speakers. The remaining 35 spoke English as an additional language.

Text. The original text was extracted from a paper about bacteriophage therapy [17]. The selection of this text was motivated by two factors: it was expected to be difficult to read because it had a high FKGL, and there was little overlap between its topic and expected student backgrounds. During the text selection process, we also tested the text to ensure that the FKGL for the simplified version (11.5) was markedly different from that of the original text (16.0) so that the two versions could be easily contrasted.

Procedures. Consistent with work by Bernholt and colleagues [4], all participants completed a pre- and post-test to assess their knowledge of the topic from the text. Participants read the text in between completing the two tests. There was no time limit, so they could engage with the material at their own pace to maximize comprehension.

Instruments. Both the pre- and post-test used a set of questions to assess learner knowledge of the topic (i.e., antibiotics, bacteriophage, and phage therapy). The tests included true/false items and multiple choice items, which have been shown to provide a dependable measure of reading comprehension [23]. The tests were designed to avoid priming effects so that performance on the post-test was not influenced by the pre-test. The pre-test contained five items and the post-test had ten items. Scores are reported as percentages.

To obtain learner perceptions of the provided texts, the questionnaire included items about the qualities of the text and whether there was missing information (Yes, Maybe, No). The students were asked to rate the comprehensibility of the text on a 5-point rating scale from "I did not understand it" (1) to "I understood it perfectly" (5). They were also asked to rate text readability from "It was hard to read" (1) to "It was easy to read".

Analyses. We present descriptive statistics of students' raw scores and ratings in addition to reporting descriptive statistics of their raw gain score (post-test − pre-test). We report this raw gain since several learners scored 100% on the pre-test, making the use of normalized learning gains infeasible for these learners.

We used single-tailed inferential statistics because we expected post-test scores to be higher than pre-test scores. Independent t-tests were used for between group comparisons when the ratio data met the assumptions of the test. A paired t-test was used when comparing pre-test to post-test performance. The Mann-Whitney test was used for rank-based data and the Chi square test was used for detecting differences in categorical data.

4.2 Results

Participant Assessment of Text Qualities. Overall, study participants reported that the texts were both easy to read and understand (Fig. 1). No differences were found between the text versions for participants' ratings of how understandable the text was ($U = 117.5, p = .143, r = .25$). However, moderate differences were found between groups for their perception of how readable the text was ($U = 91.0, p = .022, r = .40$), with the simplified text group providing higher readability scores.

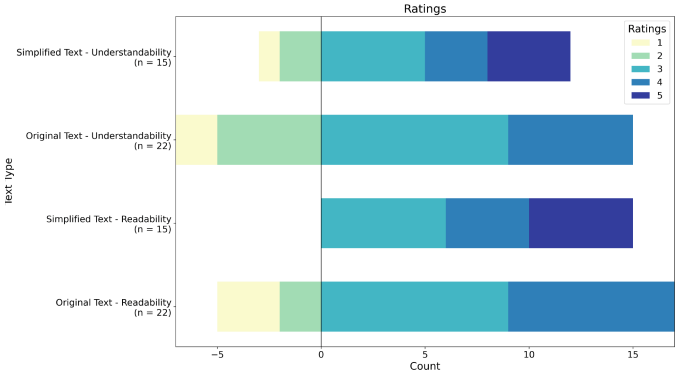


Fig. 1. Participant ratings of text comprehensibility and readability by condition

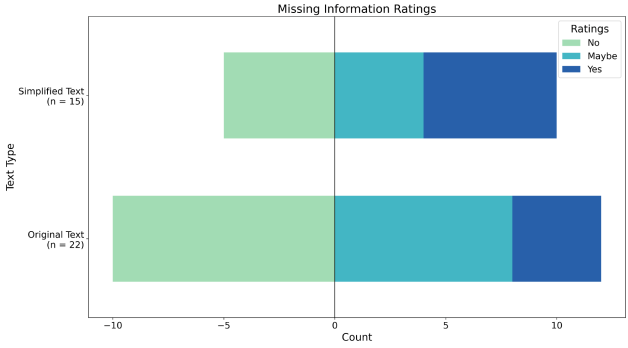


Fig. 2. Participant perceptions of whether the text was missing information they needed

There was no difference between the two conditions ($\chi^2(2) = 2.12, p = .171, \phi = .24$) for participants' assessment of whether the provided text was missing information (Fig. 2), with up to half of participants in each condition feeling that some information was missing.

Learning. A small improvement in participant scores (Fig. 3) from the pre-test ($M = 49.7, SD = 22.91$) to the post-test ($M = 57.8, SD = 16.69$) was identified across all learners: $t(36) = 1.84, p = .037, d = .31$. So those who received the simplified text learned as did those who used the original text.

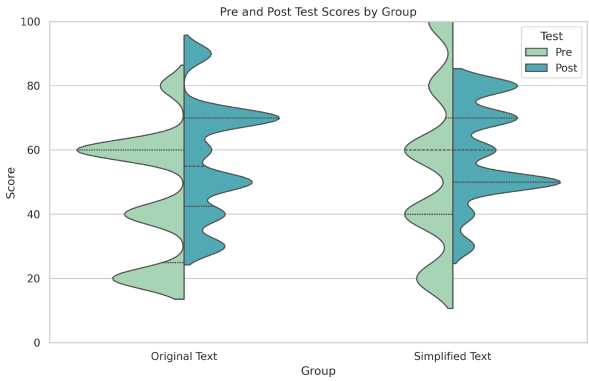


Fig. 3. Distribution of participant scores on the pre- and post-test by condition

For those in the original text condition, this meant an improvement of approximately 10% (Fig. 4) from the pre-test ($M = 46.4, SD = 19.89$) to the post-test ($M = 56.8, SD = 17.83$). For those who read the simplified texts, the observed gain from the pre-test ($M = 54.7, SD = 26.69$) to the post-test ($M = 59.3, SD = 15.34$) appeared to be smaller ($\approx 5\%$). But statistical testing found no significant differences across groups: $t(35) = 0.64, p = .263, d = .21$. It is worth noting that ceiling effects were observed on the pre-test for the simplified text group and the variability in their performance dropped from the pre- to the post-test.

5 Discussion

The complexity of academic texts poses a challenge to student success, especially for students with diverse learning needs who struggle to read. Furthermore, simplification tools carry the risk of losing key information from academic texts. So, we adopted a two-study approach to address this critique and examine the utility of LLMs in text simplification as a means of making texts more accessible to all students. Findings from both studies highlight the potential of LLMs in addressing the challenges associated with reading and comprehending academic texts.

5.1 Does Including a Readability Metric in the Prompt Support Better Text Simplification?

Study 1 demonstrated that the Metric-Guided Prompt generated academic texts with improved readability compared to those generated using the Plain-

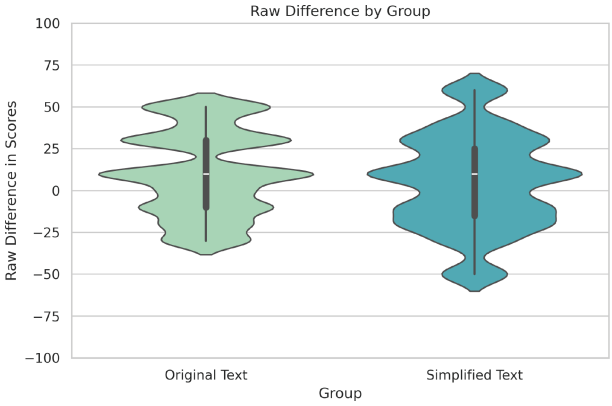


Fig. 4. Distribution of raw difference scores from the pre-test to the post-test by condition

Language Prompt. This aligns with earlier research suggesting that systematic approaches using predefined metrics can support text simplification (e.g., [11]). The guidance provided by the Metric-Guided Prompt, using the FKGL measure of readability [15], could have helped to emphasize objective readability criteria and, as such, generated simplified text that is more aligned with established standards. In contrast, the Plain-Language Prompt used a less structured approach (i.e., the LLM was prompted to rephrase sentences for clarity and replace complex vocabulary with simpler synonyms), which did not result in a change in text readability because this prompting approach preserved the variability in readability values seen in the original texts. Taken together, these results indicate the importance of including a measure of readability as part of the prompt when simplifying texts.

While the GPT-3.5 Turbo model has great capabilities, newer models with potentially better performance are being released. Further research could compare how different models influence the text simplification approach. Another limitation is the maximum token limit imposed by ChatGPT's API rules, potentially resulting in summarization instead of simplification despite the explicit prompt. Future work could address this by evaluating prompt strategies for longer academic texts outside of these constraints.

5.2 How Readable Do Students Feel the Original and Simplified Text Are?

A major criticism of readability measures (e.g., FKGL) is the focus on syllables, words, and sentence length that reflect the surface structure of a text and not its syntactic or semantic structure [36]. The findings indicate that students perceived the simplified text to be more readable than the original text. The simplified text also had a lower FKGL. Taken together, these findings from an evaluation with a single text suggest that including the readability metric when

prompting an LLM to simplify complex, academic text not only improves readability from a metric perspective but also from a learner perspective.

5.3 How Much Do Students Learn When Reading the Simplified Texts in Contrast to the Original Texts?

The students' test performance did not significantly differ between those who read the simplified text and those who read the original text, with both groups improving their scores by at least 5% on average. Although these findings should be interpreted with caution, given the small sample size and the ceiling effects observed in the pre-test of those in the simplified text condition, the comparable performance between the simplified text and original text conditions is noteworthy. Moreover, there was no difference in the amount of information students felt was missing in each condition. These results suggest LLM-simplified texts may enhance readability without compromising comprehension, preserving learning outcomes.

While the learner evaluation had a relatively small sample size, we identified important distinctions in learner perceptions. Post-hoc sample size estimation indicates that we would have needed between 527 and 1,246 more participants to reliably detect the small differences that may exist in student learning and perceptions between the original and the simplified text groups. Furthermore, we only investigated the differences between the original and simplified versions of one text on one topic. Future research is needed to validate these findings on more texts across more topics. Additionally, a larger and more diverse sample of students is needed to gain deeper insights into the interaction between the reader and the text [20]. Future work can evaluate learning based on standardized reading assessment to gain a more comprehensive understanding of the impact of LLM-simplified texts, allowing for further solidification of current knowledge on whether simplified texts could increase accessibility for diverse learners by enhancing readability while preserving comprehension.

6 Conclusions

The integration of LLM-driven tools in education offers a promising opportunity to enhance students' access to complex academic content based on their individual learning needs. These tools can simplify challenging texts, often written in non-native languages, while maintaining their essential meaning, thereby promoting educational equity. Additionally, it is possible to integrate these tools into classrooms through a teacher portal to assign students with reading difficulties simplified texts, and enroll them in support classes targeted for reading. Helping them stay on track academically while meeting specific learner reading needs, showing promise, particularly for students with diverse learning challenges, including those diagnosed with common reading and other learning disorders.

This study explored the impact of incorporating readability metrics in prompt strategies when using LLMs to improve academic text readability, gauge students' perceptions of the simplified text, and assess its impact on learning. The findings indicate that the Metric-Guided prompting approach significantly enhanced text readability, resulting in clearer and more accessible academic content. Further research is needed to refine the prompting strategies, evaluate their applicability across various text types and languages, and investigate their long-term effects on learning outcomes.

The results contribute valuable insights to the growing body of research on the application of AI in educational settings. Together, they serve as a starting point for future research aimed at refining strategic prompting, for example, by incorporating readability measures in prompts, to generate academic texts that are more accessible to support reading across diverse learner populations.

Acknowledgments. This work received financial support from Digital Futures (Sweden) and Killam Trusts (Canada).

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